Wake Early College of Information & Biotechnology
AP Physics 1
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Course Description: Physics is the study of the natural world through the principles of motion, forces, energy, waves, and fluids. This course offers an overview of foundational physics concepts and hands-on laboratory experiences, emphasizing the use of algebra to analyze real-world phenomena. Students will explore how physics principles have shaped our understanding of the universe, all within a historical and practical context. This is a college-level, preparatory course designed for students intending to pursue further education beyond high school. A cumulative final exam or project is a requirement for this course.

Prerequisites: This course involves both the theory of physical concepts and mathematical applications; therefore, students should have successfully completed Biology and be enrolled in Math III or higher.

Goals
- To assume responsibility for your own learning.
- To learn how to work productively and communicate ideas effectively in a group setting.
- To identify the relationships between physical concepts and the underlying mathematics and to learn to represent ideas and relationships verbally, graphically and mathematically.
- To develop scientific reasoning and critical thinking skills.
- To show how an understanding of physics concepts and mathematics can be used to describe, understand, and predict the workings of the world around us.

Materials
- ❏ Scientific Calculator or access to Desmos
- ❏ Three ring binder, at least 1 inch to hold:
  - ❏ Labs: Types of Graphs, Linearizing, Labs
  - ❏ 8 Units: HW and Notes
  - ❏ Paper
- ❏ Notebook paper (for binder)
- ❏ Composition Notebook
- ❏ Multi Colored Dry Erase Markers (also provided in class)
- ❏ Chromebook
- ❏ OpenStax Textbook

Topics (Timing subject to change)
1st 9 Weeks
- 1D Kinematics (13 days)
- Forces and Trans Dynamics (16 days)
- Work, Energy, Power (16 days)
- Linear Momentum (7 days)

2nd 9 Weeks
- Torque, Rotation, and Angular Momentum (10 days)

- Energy and Momentum of Rotating Systems (7 days)
- Circular Motion and Gravitation (7 days)
- Oscillations (6 days)
- Fluids (5 days)
- Review (2 days)

Grading:

Course Grading	**Quarter Grading**
40% Quarter 1	Grades will be presented on a total point system.
40% Quarter 2	***Approximate points*** of each type of assignment is as follows:
20% Final Exam	- Tests/Projects: 100 pts
	- Quizzes: 25 pts
	- Classwork/Homework: 15 pts

EXTRA HELP

It is inevitable that there will be times when you may not grasp a concept the first time. Extra help is ALWAYS available, but it is up to you to seek help as soon as possible. **Waiting until the day before or the day of the test is too late.** Here are some ways to be proactive about your studying:

1. Come in before or after school or during Maverick Time for help
2. Get together with classmates to work on assignments.
3. Email me: **hwhittington@wcpss.net**
4. Use extra materials that will consistently be provided to you.
    a. AP Classroom Videos and Topic Questions
    b. OpenStax Textbook

Assessments: Formative quizzes will be assigned regularly. I do not expect students to ace every quiz…they are helping me assess how you learn! Corrections or retakes will always be offered. I calculate and post grades as a percentage. I will provide verbal and written feedback on assignments when turned in by hand and virtually when turned in via Canvas.

Retests: Retests will be given to any student who scores below at 80% on an assessment. To qualify for retesting, corrections must be done on the missed test questions and students must visit me during Maverick time prior to the retest to go over the last assessment. An alternate assessment or portions of the assessment (if only certain topics are missed) will be given at a later date. I encourage students to get retests done within a week of the graded test being returned.

Final Exam: All courses will include a comprehensive written final examination or project that will count for 20% of the final grade. All final examinations will occur during the school-wide scheduled final exam window.

Class Expectations: All students are expected to adhere to all Wake County and WECIB rules and policies. All students should attend class meetings as scheduled, participate in all class activities, complete all work, and submit assignments in a timely manner.

Canvas: Assignments will consist of a mix of online and paper assignments. All assignments will be posted on Canvas for students to view and submit as needed. Please view my Canvas page here:
Project Based Learning: I aim to assign and complete at least two major projects throughout the semester. Details, deadline, rubrics, and more will be provided at the beginning of the project and posted on Canvas.

Communication: I respond well to emails and will usually message back within the same day if during the school week. I like to email with positive comments on the students as well as answer any questions you may have throughout the semester. I will also reach out through Talking Points and encourage you to do so as well. I will send a test TalkingPoints at the beginning of the semester.

Availability: I am available in the mornings from 6:55-7:15 during my lobby duty for general help and questions. I will also be available during Maverick Time at lunch hours and may contact you for needed tutoring or assignment completion. I can be available after school as needed if you check with me ahead of time and have a secured ride after school.

Familial Support: I will send email blasts and TalkingPoint blasts to students and parents throughout the semester with events surrounding what we are learning in class, novel recommendations, extra curricular activities, etc. Be on the lookout! If you have any suggestions, please email or send me a message as well!

Digital Citizenship and Electronic Devices

- Students are expected to recognize that electronic devices should “provide opportunities to enhance learning and improve communication within the school community and with the larger global community” (WCPSS Board Policy 3225).

Academic Honesty/Honor Code

- Students and staff are expected to follow WCPSS’s Honor Code.
- Teachers are expected to make parent/guardian contact for academic honesty violations.
- Students are expected to complete the alternative replacement assignment of equivalent rigor with an academic deduction of having the lowest possible letter grade of the earned grade. Ex) If a student scores an 88 on the alternative assignment, an 80 will be placed in the gradebook.

Campus and Hallway Expectations (during class changes and class time)

- Students are expected to have a visible bathroom/hall pass when in the hallway during class time.
- Students are expected to respect the learning environment and remain quiet in the hallway during class time.
- Students are expected to adhere to and follow posted restricted and non-restricted areas of the campus at all times, including lunch.
- Students will not leave the classroom during the first or last 10 minutes of the class period.

Tentative Pacing Guide for Fall 2026

Week of	Monday	Tuesday	Wednesday	Thursday	Friday
August 3-7				Unit 1a: Kinematics 1D	**ACT Prep**
August 10-14	U1a	U1a	**ACT Prep**	**ACT Prep Kinematics, motion maps**	U1a Free fall
August 17-21	U1a Quiz Free fall lab	U1a Finish/Revie w start U1b	U1a Assessment	U1b	U1b
August 24-28	U1b	U1b Assessment	Unit 2: Forces and Dynamics	U2	U2
August 31 - September 4	U2	U2	U2	U2	U2
September 7-11	**Holiday**	U2	U2	U2 Assessment	Unit 3: Work, Energy, Power
September 14-18	U3	U3	U3	U3	U3
September 21-25	**Teacher Work Day**	U3 Assessment	Unit 4: Work & Energy	U4	U4
September 28 - October 2	U4	U4	U4	U4	U4
October 5-9	**Teacher Work Day**	**Teacher Work Day**	U4	U4	U4

October 12-16	U4 Assessment	U4 Flex Day/ Catch-Up/Corrections	**PSAT**	Unit 5: Momentum & Impulse	U5
October 19-23	U5	U5	**PACT**	U5	**Fall Festival** U5 Assessment
October 26-30	Field Trip Prep	Field Trip	Workday	Workday	Presentations
November 2-6	Unit 6 Fluids	**Teacher Work Day**	U6	U6	U6
November 9-13	U6	**Teacher Work Day**	**Holiday**	U6	U6
November 16-20	U6 Assessment	Unit 7 Rotation	U7	U7	U7
November 23-27	U7	U7 Assessment	**Holiday**	**Holiday**	**Holiday**
November 30 - December 4	Unit 8a: Circular Motion	U8a	U8a	U8b: Harmonics and Gravitation	U8b
December 7-11	U8b	U8b	U8b Assessment	Review	Final Exam
December 14-18	**Exams**	**Exams**	**Exams**	**Exams**	**Exams**

AP Physics 1 Testing Specifics

2025 - 2026 Test Format and Grading

In order to prepare you for the AP exam in May, I have chosen to format and grade each test, as close as possible, to match the actual AP exam experience.

Test Format:
The format of the AP test includes a multiple choice section and a Free Response section, each worth 50% of the total score. The number of questions and the given amount of time on the AP exam requires the use of time management. Here is the College Board's recommendation for time management on the AP exam: "...if they (students) do not properly budget their time, they may have insufficient time to complete all the multiple-choice questions in Section I and all of the free-response questions in Section II. Students often benefit from taking a practice exam under timed conditions prior to the actual administration." For this reason, the difficulty and number of multiple choice questions and free response questions are chosen to attempt to match the difficulty and pacing of the actual AP exam. I want to give you many opportunities to practice pacing yourself while working through AP level questions.

AP Physics 1 Student Name ______________________________ Block__________

CODE OF CONDUCT

- Respect yourself, others, and school property.
- Come to class prepared. This means always having your CHARGED chromebook, binder with paper in it, something to write with and your completed homework assignment. *Pencils and calculators will be required on all test days.*
- Be in your assigned seat before the bell rings and wait to be dismissed by the bell or the teacher at the end of class. *Anyone entering after the bell without a pass will be considered tardy*.
- Follow all lab safety rules outlined by the teacher.
- Follow all school policies in this classroom. ** (Consult the student handbook for the school policy on cheating, dress code, and food and drink)**
- **CELL PHONES OR OTHER ELECTRONIC DEVICES SHOULD BE LIMITED IN CLASS *UNLESS SPECIFICALLY ALLOWED BY THE TEACHER*.**

LAB SAFETY RULES

1. Follow all directions. This is very important while engaging in lab activities.
2. Perform only authorized investigation. When creating your own lab procedures, always have them approved by the teacher before you begin.

3. Be careful of loose clothing and hair. This could be dangerous during some experiments.
4. Wear protective equipment whenever directed. If you fail to do so you will be asked to leave the lab.
5. Know emergency procedures and where you may find the necessary equipment.
6. All accidents and unusual occurrences must be reported to the teacher immediately. Do not panic and do not clean up an accident without the approval of the teacher.
7. No horseplay in the science lab. This may be one of the least safe places for such behavior.
8. If you don’t understand something, please ask.
9. All broken glass must be placed in a separate container which is labeled for that purpose, and other waste must be disposed of properly.

I have read and understand the class grading policy, AP test format and grading, class code of conduct, and the lab safety rules.   I agree to follow all rules and policies outlined for this class.

Student signature ____ Date______________

Parent/Guardian signature _________________________________ Date______________